

Mini Project (CSS7102)
Review-0 (11th August 2026)

PRJ-465 Emergency Shelter Allocation Platform

Group Members	
Sharan (Team Leader)	20231ISE0089
Suhas S Kattimani	20231ISE0090
Harshil Gurjar	20231ISE0056

Under the supervision of
Mr. Irfan Rajab Bhat
[Assistant Professor / Associate Professor / Professor]
Department of Computer Science and Engineering

Abstract

Access to safe, well-organized emergency shelter is a critical component of effective disaster response, directly aligned with UN Sustainable Development Goal 11 (Sustainable Cities and Communities). Traditional shelter allocation during disasters — such as the recurring floods across Assam and other high-risk regions of India — relies on manual, ad hoc decisions by field officials, which are slow, inconsistent, and often leave some shelters overcrowded while others sit under-utilized. This project proposes an AI-enabled Emergency Shelter Allocation Platform that leverages optimization and data analytics techniques to compute an optimal shelter allocation plan given data on shelter capacities and displaced populations.

The system models shelters (location, capacity) and affected zones (population, location, vulnerability level) as structured data, then formulates the allocation problem as a linear program that minimizes overcrowding and total travel distance while respecting capacity constraints. A prioritization mechanism ensures that vulnerable groups — elderly, children, and persons with disabilities — are given precedence whenever total demand exceeds available shelter capacity. The project is implemented using Python-based tools (pandas for data handling, PuLP/SciPy for optimization) and is evaluated on synthetic disaster scenarios modeled on real-world flood and cyclone events, benchmarked against a naive/manual allocation baseline.

The final deliverable includes an interactive Streamlit-based dashboard where disaster management authorities can input shelter and population data to receive an optimal allocation plan, along with visualizations of shelter utilization and unmet demand. By combining proven optimization techniques with an accessible, low-cost web interface, this project aims to deliver a scalable decision-support tool for emergency shelter allocation — reducing response time, improving fairness for vulnerable populations, and requiring no specialized hardware or infrastructure.